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Financial Parallelverse

Complete System Execution & Demonstration Manual — Explore your financial futures before making a decision.

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1. Executive Summary & Problem Solved

Standard EMI calculators only compute flat repayments based on static assumptions. They fail to warn users what happens when real life strikes: a delayed paycheck, sudden hospital bills, or unexpected inflation. **Financial Parallelverse** solves this by running **5 concurrent deterministic simulations**, identifying financial breaking points, comparing purchase pathways (Cash vs. Save-and-Wait vs. Cheaper Alternative vs. EMI), and generating downloadable **iCalendar (.ics) reminders** for scheduled installments.

Component	Technology	Port / Location	Role
FastAPI Backend	Python 3.13 / Uvicorn	http://localhost:8000	REST API & Simulation Logic
Interactive Dashboard	Streamlit + Plotly	http://localhost:8501	Primary Fintech User Interface
Database Engine	SQLite / SQLAlchemy 2.x	data/financial_parallelverse.db	Profile & Installment Persistence
Test Suite	Pytest	23 Automated Test Cases	Engine & API Reconciliation

2. Step-by-Step Execution Order

Follow these steps sequentially to launch and verify the complete application:

Step 1: Open the Project Directory

Open PowerShell or Command Prompt and navigate to the project directory on your desktop:

```
cd "C:\Users\Raghoothama T R\Desktop\Financial_Parallelverse"
```

Step 2: Seed Demo Database

Populate the SQLite database with realistic profile values and initial EMI installment plans:

```
python scripts/seed_demo.py
```

Step 3: Run the 23-Test Verification Suite

Execute the comprehensive Pytest suite to mathematically verify all financial and EMI algorithms:

```
python -m pytest backend/tests -v
# Or double-click: run_tests.bat
```

Step 4: Launch the FastAPI Backend Service

Start the FastAPI application. Access OpenAPI documentation at http://localhost:8000/api/docs:

```
python -m uvicorn backend.app.main:app --host 0.0.0.0 --port 8000 --reload
# Or double-click: run_backend.bat
```

Step 5: Launch the Streamlit FinTech Application

Open a second terminal window and start the primary Streamlit frontend:

```
python -m streamlit run frontend_streamlit/app.py --server.port 8501
# Or double-click: run_streamlit.bat
```

ONE-CLICK LAUNCHER (Alternative)

To automatically launch both the backend and frontend in separate dedicated windows, run:

```
run_all.bat
```

3. Feature Walkthrough & Hackathon Demonstration Guide

Use this sequence during your presentation to demonstrate the novelty and polish of the application to the judges:

1. Overview Dashboard

Displays key financial vitals: Monthly Free Cash Flow, Current Savings, Emergency Runway (Months), and Debt-to-Income (DTI). Features an interactive inflow/outflow donut chart showing exactly where monthly income goes.

2. Parallel Futures Simulator (Core Innovation)

Simulates 5 parallel timelines across a 6–24 month horizon:
• **Baseline Reality**: Normal conditions.
• **Emergency Medical Shock**: Sudden ■25,000 out-of-pocket expense in Month 3.
• **Delayed Salary Flow**: Timing shift where a paycheck is delayed for 1 month.
• **Cost of Living Inflation**: 15% increase in monthly essential & discretionary expenses.
• **Parallel Nightmare**: Triple adverse shock.
Visualized using interactive multi-line Plotly graphs with the minimum emergency buffer line and automated breaking-point alerts.

3. Purchase Affordability Decision Matrix

Evaluates acquiring a ■70,000 laptop via 4 pathways: Buy Now with Cash, Save First & Buy Later, Buy a Cheaper Alternative (■42,000), or Buy via EMI. Shows safety ratings, emergency buffer impact, and trade-off analysis.

4. Reducing-Balance EMI & Amortization Calculator

Computes exact EMI using standard mathematical formulas. Reconciles monthly interest and principal down to ■0.00 final balance, supporting zero-interest loans and early prepayments.

5. Installment Tracker & One-Click Calendar (.ICS) Reminders

Displays active EMI plans with progress bars, due dates, and overdue status detection. Features a **Download Calendar (.ICS) Reminders** button to import reminders directly into Google Calendar, Apple Calendar, or Outlook.

6. Financial Profile Persistence

Allows editing and saving monthly income, expenses, debts, and emergency targets with instant persistence to SQLite.

4. Backend REST API Contract

Method	Endpoint	Description
GET	/api/health	Health check & service status
GET / PUT	/api/profile	Retrieve and update user financial profile
POST	/api/scenarios/simulate	Execute 5-timeline parallel future simulation
POST	/api/purchases/evaluate	Evaluate 4-way purchase affordability options
POST	/api/emi/calculate	Calculate reducing-balance loan amortization
GET / POST	/api/installments	List all plans or create a new installment plan
PATCH	/api/installments/{id}/payments/{pid}	Mark installment payment as PAID or PENDING
GET	/api/installments/{id}/calendar	Generate and download .ics calendar reminder file

5. Quality Assurance & Test Verification

The application has been tested against 23 automated unit and integration tests covering: income-to-expense ratios, cash deficit breaking points, emergency fund depletion, zero-interest loans, reducing-balance mathematical reconciliation, month-end date clamping (Feb 28, Apr 30), and .ics calendar generation. **Result: 23 passed, 0 failed.**